

RESEARCH ARTICLE OPEN ACCESS

Recursive Feasibility of Nonlinear Stochastic Model Predictive Control With Gaussian Process Dynamics

Daniel Landgraf | Andreas Völz  | Knut Graichen

Chair of Automatic Control, Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen, Germany

Correspondence: Andreas Völz (andreas.voelz@fau.de)**Received:** 16 September 2025 | **Revised:** 16 January 2026 | **Accepted:** 14 February 2026**Keywords:** Gaussian processes | recursive feasibility | stabilizing conditions | stochastic model predictive control

ABSTRACT

Data-based learning of system dynamics allows model-based control approaches to be applied to systems with partially unknown dynamics. Gaussian process regression is a preferred approach that outputs not only the learned system model but also the variance of the model, which can be seen as a measure of uncertainty. Stochastic model predictive control uses a stochastic system model like this to propagate the probability density functions of the predicted states and solves a stochastic optimal control problem in every time step. This paper proposes a stochastic model predictive control algorithm with guarantees on recursive feasibility and stability for nonlinear systems with unknown parts that are learned from data using Gaussian process regression. To this end, an error bound of the Gaussian process prediction is established based on an upper bound of the norm of the unknown dynamics function in the corresponding reproducing kernel Hilbert space. Unlike related work, this allows to consider state and input dependent disturbances instead of independent and identically distributed (iid) white Gaussian noise which is usually considered in the context of stochastic model predictive control. The numerical evaluation illustrates that the resulting control algorithm stabilizes nonlinear systems with partially unknown dynamics.

1 | Introduction

Model predictive control (MPC) represents a powerful method for controlling constrained nonlinear systems by solving an optimal control problem (OCP) online [1, 2]. As shown in [3], under certain assumptions, a model predictive controller stabilizes a system even if there are small deviations between the system model and the true system dynamics. However, if the difference between the analytical model and actual behavior is large, it is desirable to learn this difference to improve the system model. Possible learning methods, which can be applied in the context of MPC, are Gaussian processes (GPs), see for example [4, 5], and neural networks, see for example, [6, 7].

In this article, we focus on the modeling of unknown dynamics with GPs because they allow to evaluate the variance of the

model and the model error can be bounded under certain conditions [8–10]. However, the assumptions are strict and difficult to evaluate in practice. A less restrictive approach is proposed in [11] using the Lipschitz continuity of the unknown system function. It states a probabilistic uniform error bound which, due to its probabilistic nature, is not suitable for proving recursive feasibility if hard state constraints must be taken into account. However, a deterministic error bound can be found if the unknown function is an element of a reproducing kernel Hilbert space with bounded norm [12].

The evaluation of the model variance allows to predict not only the mean trajectory of the state but also higher stochastic moments of the predicted state. Stochastic model predictive control (SMPC) considers stochastic properties of the predicted state, such as the probability density function or a set of

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *International Journal of Robust and Nonlinear Control* published by John Wiley & Sons Ltd.

stochastic moments, and solves an OCP in every time step, see [13] for an overview. This allows for a more realistic description of the system, as model uncertainties and measurement noise can be taken into account. It is shown in [14] that guarantees on stability of the closed-loop system can be derived for linear systems. Furthermore, recursive feasibility can be guaranteed if the support of the process noise is bounded and convex [15]. These papers, however, assume that the system function and the distribution of the disturbance is known. If this is not the case, behavioral systems theory can be applied to linear SMPC, for which exponential stability of the closed-loop system and recursive feasibility can be shown [16]. Similar results can be found in [17] for a linear SMPC approach that utilizes polynomial chaos expansion to calculate the stochastic moments of the predicted state.

For nonlinear systems, the stability analysis is more complicated, mainly because of the non-trivial uncertainty propagation, see [18] for a discussion. An early approach for nonlinear systems was proposed in [19], which shows stability based on a global stochastic Lyapunov function if the system dynamics and the distribution of the disturbance are known. However, this work does not take constraints into account, which limits its applicability. If state constraints must be considered, [20] derives guarantees on stability and recursive feasibility assuming that the disturbance takes values in a compact set and the system function is known exactly. Since the evaluation of a GP is always a Gaussian distributed random variable, the set of possible disturbances is not compact in this case, for which reason this approach cannot be used directly in combination with GPs. Alternatively, control Lyapunov-barrier functions [21] or the concept of robust asymptotic stability in expectation [22, 23] can be used. Moreover, probabilistic reachable sets can be applied to SMPC in order to tighten the constraints such that closed-loop guarantees can be found, even if the disturbance is unbounded [24]. However, knowledge of the exact system dynamics and stochastic properties of the disturbance is a major obstacle to the application of these methods. Even though [25] shows that SMPC is inherently robust to sufficiently small errors of the disturbance distribution or the system model, this result does not hold for larger deviations between the system model and the real dynamics.

The main contribution of this paper is the proposition of a data-driven SMPC algorithm for nonlinear systems with guaranteed recursive feasibility. To this end, the unknown parts of the system dynamics are learned using Gaussian process regression (GPR). We show that an error bound for the GP predictions can be found under mild conditions and present a reformulated OCP which can be solved with standard solvers and for which the desired guarantees are proven.

The remainder of this article is structured as follows. Section 2 states the considered problem including the system class and some basic assumptions. In Section 3, GP regression is introduced and error bounds for the GP predictions are derived. In Section 4, the OCP of the stochastic model predictive controller is defined, constraint satisfaction and recursive feasibility is proven, and it is shown that the cost decreases in expectation. Finally, a numerical example is studied in Section 5 and the main aspects of the paper are summarized in Section 6.

Notation: A vector $\mathbf{a}$ is written in bold and a matrix $\mathbf{A}$ is denoted by a bold capital. The sets $\mathbb{I}$ and $\mathbb{R}$ denote the integers and the reals. The matrix $\mathbf{A} = \text{diag}(\mathbf{a})$ denotes the diagonal matrix with diagonal elements equal to the elements of the vector $\mathbf{a}$ and $\text{tr}(\mathbf{B})$ denotes the trace of a matrix $\mathbf{B}$. The norm $\|\mathbf{a}\|$ of a vector $\mathbf{a}$ is defined as the Euclidean norm. The symbol for a random variable X is a non-bold capital, irrespective of whether it is a vector or a scalar, and its expected value is denoted by $\mathbb{E}[X]$. For a scalar random variable X , the variance is $\text{Var}[X]$, while the covariance of two scalar random variables X and Y is $\text{Cov}[X, Y]$. The covariance matrix of a vector-valued random variable X is denoted by $\text{Cov}[X]$. A random variable X which is normally distributed with mean $\mathbf{a}$ and covariance matrix $\mathbf{B}$ is written as $X \sim \mathcal{N}(\mathbf{a}, \mathbf{B})$. A function $\alpha : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ belongs to class $\mathcal{K}_{\infty}$ if it is continuous, strictly increasing, and $\alpha(0) = 0$.

2 | Problem Statement

A nonlinear discrete-time system

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{g}(\mathbf{x}_k, \mathbf{u}_k) \quad (1)$$

is considered with state $\mathbf{x} \in \mathbb{X} \subseteq \mathbb{R}^{N_x}$ and control input $\mathbf{u} \in \mathbb{U} \subseteq \mathbb{R}^{N_u}$, where $\mathbb{X}$ and $\mathbb{U}$ are non-empty measurable sets for the states and control inputs, respectively. The first part of the system dynamics is described by the Borel-measurable function $\mathbf{f} : \mathbb{X} \times \mathbb{U} \rightarrow \mathbb{X}$ and is assumed to be known such that it can be used directly in the algorithm of the controller. The Borel-measurable function $\mathbf{g} : \mathbb{X} \times \mathbb{U} \rightarrow \mathbb{X}$ represents an unknown part of the system dynamics and is learned from data points using GPR. The control input is chosen as

$$\mathbf{u}_k = \boldsymbol{\pi}(\mathbf{x}_k, \mathbf{v}_k), \quad (2)$$

where $\boldsymbol{\pi} : \mathbb{X} \times \mathbb{V} \rightarrow \mathbb{U}$ denotes a Borel-measurable control policy with parameters $\mathbf{v}_k \in \mathbb{V} \subseteq \mathbb{R}^{N_v}$ and a non-empty measurable set $\mathbb{V}$. A common choice is an affine function of the states

$$\boldsymbol{\pi}(\mathbf{x}_k, \mathbf{v}_k) = \mathbf{v}_{1,k} \mathbf{x}_k + \mathbf{v}_{2,k} \quad (3)$$

with $\mathbf{v}_{1,k} \in \mathbb{R}^{N_u \times N_x}$, $\mathbf{v}_{2,k} \in \mathbb{R}^{N_u}$, and $\mathbf{v}_k$ is the vector of all elements of $\mathbf{v}_{1,k}$ and $\mathbf{v}_{2,k}$, see for example [14]. The choice of the parameter $\mathbf{v}_k$ as an optimization variable is called closed-loop prediction MPC and can prevent overly conservative results of a model predictive controller [26]. Note, however, that (2) allows the parameterization $\mathbf{u}_k = \mathbf{v}_k$ for which an optimization over $\mathbf{v}_k$ is equivalent to an optimization over $\mathbf{u}_k$. In order to simplify the notation, the resulting closed-loop system is referred to as

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{f}(\mathbf{x}_k, \boldsymbol{\pi}(\mathbf{x}_k, \mathbf{v}_k)) + \mathbf{g}(\mathbf{x}_k, \boldsymbol{\pi}(\mathbf{x}_k, \mathbf{v}_k)) \\ &=: \mathbf{f}_{\pi}(\mathbf{z}_k) + \mathbf{g}_{\pi}(\mathbf{z}_k), \end{aligned} \quad (4)$$

where $\mathbf{z} = (\mathbf{x}, \mathbf{v}) \in \mathbb{Z}$ with $\mathbb{Z} = \mathbb{X} \times \mathbb{V}$. Furthermore, it is assumed that the system must satisfy the constraint

$$\mathbf{z} \in \mathbb{Z} = \{(\mathbf{x}, \mathbf{v}) \in \mathbb{X} \times \mathbb{V} \mid h_j(\mathbf{x}, \mathbf{v}) \leq 0 \quad \forall j \in \mathbb{I}_{[1, N_h]}\} \quad (5)$$

with Borel-measurable constraint functions $h_j : \mathbb{X} \times \mathbb{V} \rightarrow \mathbb{R}$, where N_h is the number of constraint functions. Note that pure state constraints and pure input constraints are a special case of (5) if the control policy $\boldsymbol{\pi}$ is known.

The function g_π may not be arbitrarily complex because this would make it impossible to state guarantees regarding recursive feasibility of a controller if constraints must be considered. For this reason, the following assumption restricts the class of functions for the elements of g_π to functions from reproducing kernel Hilbert spaces (RKHS) with bounded norm.

Assumption 1. The functions f_π and g_π are continuous and satisfy $f_\pi(\mathbf{0}) = \mathbf{0}$ and $g_\pi(\mathbf{0}) = \mathbf{0}$. Furthermore, there exist RKHS $\mathcal{H}_{k,i}$ for all $i \in \llbracket 1, N_x \rrbracket$ with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}_{k,i}}$ and corresponding positive definite, continuously differentiable, and bounded kernels k_i such that $g_{\pi,i} \in \mathcal{H}_{k,i}$ with bounded norm $\|g_{\pi,i}\|_{\mathcal{H}_{k,i}} := \sqrt{\langle g_{\pi,i}, g_{\pi,i} \rangle_{\mathcal{H}_{k,i}}} \leq b_i$ for some $b_i \geq 0$. The set of functions which fulfill this assumption is denoted by $\mathcal{G}$.

The restriction to functions $g_\pi \in \mathcal{G}$ allows to infer Lipschitz continuity of the function as shown in the following lemma.

Lemma 1. Let Assumption 1 hold. Then, $g_\pi \in \mathcal{G}$ is Lipschitz continuous with respect to $\mathbf{x}$ such that

$$\|g_\pi(\mathbf{z}) - g_\pi(\mathbf{z}')\| \leq L_g \|\mathbf{x} - \mathbf{x}'\| \quad (6)$$

holds for all $\mathbf{z} = (\mathbf{x}, \mathbf{v}) \in \mathcal{Z}$, $\mathbf{z}' = (\mathbf{x}', \mathbf{v}) \in \mathcal{Z}$, where $L_g = \sqrt{2b_{\max}k'_{\max}}$ is the Lipschitz constant of g_π with

$$b_{\max} = \max_{i \in \llbracket 1, N_x \rrbracket} b_i \quad (7)$$

$$k_{\max} = \max_{i \in \llbracket 1, N_x \rrbracket} \|k_i\|_{\infty} \quad (8)$$

$$k'_{\max} = \max_{i \in \llbracket 1, N_x \rrbracket} \left\| \frac{\partial k_i}{\partial \mathbf{x}} \right\|_{\infty}. \quad (9)$$

Proof. See [27, Lemma 2]. $\square$

The Lipschitz continuity of g_π is required in Section 3.2 in order to bound the error of GP predictions. Furthermore, Lipschitz continuity is assumed for the known part of the system dynamics and the constraint functions.

Assumption 2. The functions f_π and h_j , $j \in \llbracket 1, N_h \rrbracket$ are Lipschitz continuous with respect to $\mathbf{x}$ such that

$$\|f_\pi(\mathbf{z}) - f_\pi(\mathbf{z}')\| \leq L_f \|\mathbf{x} - \mathbf{x}'\| \quad (10)$$

$$|h_j(\mathbf{x}, \mathbf{v}) - h_j(\mathbf{x}', \mathbf{v})| \leq L_{h,j} \|\mathbf{x} - \mathbf{x}'\| \quad (11)$$

holds for all $\mathbf{z} = (\mathbf{x}, \mathbf{v}) \in \mathcal{Z}$, $\mathbf{z}' = (\mathbf{x}', \mathbf{v}) \in \mathcal{Z}$, where L_f and $L_{h,j}$ are the Lipschitz constants of f and h_j .

3 | GPs

As the system dynamics function g_π in (4) is assumed to be unknown, it cannot be used directly by the controller. However, data points can be collected in order to build a regression model that approximates the unknown function g_π . GPR is a particularly suitable regression method in conjunction with

SMPC because it provides not only a model for the most likely outcome of the unknown function but also a measure of uncertainty. For this reason, GPR is introduced in the following and error bounds of GP predictions are derived, which are necessary to prove recursive feasibility of the stochastic model predictive controller.

3.1 | GPR

GPR is a machine learning method that can be used to approximate functions based on prior assumptions and collected data, see [28] for details. In this work, it is used to find an approximation of the unknown function g_π . The regression model is denoted by $\tilde{g}_\pi$ in order to distinguish between the model and the true dynamics function. Each element of $\tilde{g}_\pi$ is an independent GP

$$\tilde{g}_{\pi,j} \sim \mathcal{GP}(m_j, k_j) \quad j \in \llbracket 1, N_x \rrbracket, \quad (12)$$

where $m_j : \mathcal{Z} \rightarrow \mathbb{R}$ and $k_j : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}$ are the corresponding mean and covariance functions, respectively. Since prior knowledge is already taken into account by the function f_π , the mean functions are assumed to be zero in the following. Each kernel k_j represents the prior knowledge about the corresponding element of the unknown function g_π and defines the corresponding RKHS.

It is assumed that M input data points $\mathbf{z}_l^{\text{in}}$ and output data points $\mathbf{z}_l^{\text{out}} = g_\pi(\mathbf{z}_l^{\text{in}})$ with $l \in \llbracket 1, M \rrbracket$ are available. Then, the posterior distribution of the regression model at a point $\mathbf{z} \in \mathcal{Z}$ is given as a multivariate Gaussian distribution

$$\tilde{g}_\pi(\mathbf{z}) \sim \mathcal{N}(\boldsymbol{\mu}(\mathbf{z}), \boldsymbol{\Sigma}(\mathbf{z})) \quad (13)$$

with mean $\boldsymbol{\mu}(\mathbf{z}) = [\mu_1(\mathbf{z}), \dots, \mu_{N_x}(\mathbf{z})]^\top$ and covariance matrix $\boldsymbol{\Sigma}(\mathbf{z}) = \text{diag}([\sigma_1^2(\mathbf{z}), \dots, \sigma_{N_x}^2(\mathbf{z})]^\top)$. Let $\mathbf{Z}^{\text{out}} = [\mathbf{z}_1^{\text{out}}, \dots, \mathbf{z}_M^{\text{out}}]^\top \in \mathbb{R}^{M \times N_x}$ be the matrix of output data points. The mean and variance of the posterior elements at point $\mathbf{z}$ can be calculated for $j \in \llbracket 1, N_x \rrbracket$ by [28]

$$\mu_j(\mathbf{z}) = \mathbb{E}[\tilde{g}_{\pi,j}(\mathbf{z})] = \mathbf{k}_j^*(\mathbf{z})^\top \mathbf{K}_j^{-1} \mathbf{Z}_j^{\text{out}} \quad (14)$$

$$\sigma_j^2(\mathbf{z}) = \text{Var}[\tilde{g}_{\pi,j}(\mathbf{z})] = k_j(\mathbf{z}, \mathbf{z}) - \mathbf{k}_j^*(\mathbf{z})^\top \mathbf{K}_j^{-1} \mathbf{k}_j^*(\mathbf{z}), \quad (15)$$

where $\mathbf{Z}_j^{\text{out}}$ is the j th column of $\mathbf{Z}^{\text{out}}$, $[\mathbf{K}_j]_{ll'} = k_j(\mathbf{z}_l^{\text{in}}, \mathbf{z}_{l'}^{\text{in}})$, and $[\mathbf{k}_j^*(\mathbf{z})]_l = k_j(\mathbf{z}, \mathbf{z}_l^{\text{in}})$ with $l, l' \in \llbracket 1, M \rrbracket$. The computation time that is required to calculate (14) and (15) increases cubically with M [28]. It is assumed that the data points and the resulting mean function fulfill the following assumptions.

Assumption 3. The data points $\mathbf{z}_l^{\text{out}}$ with $l \in \llbracket 1, M \rrbracket$ are collected from noise-free observations, and the matrices $\mathbf{K}_j$ are invertible for all $j \in \llbracket 1, N_x \rrbracket$.

Assumption 4. The function $\boldsymbol{\mu}(\mathbf{z})$ is Lipschitz continuous with respect to $\mathbf{x}$ such that

$$\|\boldsymbol{\mu}(\mathbf{z}) - \boldsymbol{\mu}(\mathbf{z}')\| \leq L_\mu \|\mathbf{x} - \mathbf{x}'\| \quad (16)$$

holds for all $\mathbf{z} = (\mathbf{x}, \mathbf{v}) \in \mathcal{Z}$, $\mathbf{z}' = (\mathbf{x}', \mathbf{v}) \in \mathcal{Z}$, where L_μ is the Lipschitz constant of $\boldsymbol{\mu}$.

Remark 1. Assumption 4 does not lead to significant restrictions in most cases. If the function g_π and the kernels $k_i, i \in \llbracket 1, N_x \rrbracket$ are Lipschitz, then μ is also Lipschitz and an upper bound of the Lipschitz constant can be calculated [11, Theorem 3.1].

In general, SMPC requires not only a prediction of the next state but also a prediction of the following states. For multi-step predictions with a GP, the search point $\mathbf{z}$ itself is not a deterministic point but a random variable that depends on the output of the previous prediction step. Therefore, we introduce the following notation. The variables $X_{i|k}$ and $Z_{i|k}$ denote the random variables of the predicted $\mathbf{x}$ and $\mathbf{z}$, respectively, for $i \geq 0$ prediction steps ahead of the current time step k if the control policy $\mathbf{v}_{i|k}$ is applied to the system. The probability density function of the predicted state can be calculated iteratively using the law of total probability [29]

$$p(X_{i+1|k}) = \int_{\mathcal{X}} p(\mathbf{f}_\pi(Z_{i|k}) + \tilde{\mathbf{g}}_\pi(Z_{i|k}) | X_{i|k}) p(X_{i|k}) dX_{i|k}, \quad (17)$$

where $p(\mathbf{f}_\pi(Z_{i|k}) + \tilde{\mathbf{g}}_\pi(Z_{i|k}) | X_{i|k})$ is the conditional probability distribution of $X_{i+1|k}$ given a value of $X_{i|k}$. The integral (17) is difficult to evaluate for general nonlinear functions. However, the mean and the covariance matrix of the state X_{k+1} can be evaluated as shown in the following lemma.

Lemma 2. Let Assumption 1 hold. Then, the mean and the covariance matrix of the predicted state $X_{i+1|k}$ are given by

$$\mathbb{E}[X_{i+1|k}] = \mathbb{E}[\mathbf{f}_\pi(Z_{i|k})] + \mathbb{E}[\mu(Z_{i|k})] \quad (18)$$

$$\text{Cov}[X_{i+1|k}] = \text{Cov}[\mathbf{f}_\pi(Z_{i|k}) + \mu(Z_{i|k})] + \mathbb{E}[\Sigma(Z_{i|k})] \quad (19)$$

for all $i \geq 0$.

Proof. Similarly to [29], the law of total expectation can be applied, which results in

$$\begin{aligned} \mathbb{E}[X_{i+1|k}] &= \mathbb{E}[\mathbb{E}[X_{i+1|k} | X_{i|k}]] = \mathbb{E}[\mathbb{E}[\mathbf{f}_\pi(Z_{i|k}) + \tilde{\mathbf{g}}_\pi(Z_{i|k}) | X_{i|k}]] \\ &= \mathbb{E}[\mathbb{E}[\mathbf{f}_\pi(Z_{i|k}) | X_{i|k}]] + \mathbb{E}[\mathbb{E}[\tilde{\mathbf{g}}_\pi(Z_{i|k}) | X_{i|k}]] \\ &= \mathbb{E}[\mathbf{f}_\pi(Z_{i|k})] + \mathbb{E}[\mu(Z_{i|k})] \end{aligned}$$

for the mean. The m th and the n th element of $X_{i+1|k}$ are denoted by $X_{i+1|k,m}$ and $X_{i+1|k,n}$, respectively, and their covariance can be calculated using the law of total covariance by

$$\begin{aligned} \text{Cov}[X_{i+1|k,m}, X_{i+1|k,n}] &= \mathbb{E}[\text{Cov}[X_{i+1|k,m}, X_{i+1|k,n} | X_{i|k}]] \\ &\quad + \text{Cov}[\mathbb{E}[X_{i+1|k,m} | X_{i|k}], \mathbb{E}[X_{i+1|k,n} | X_{i|k}]] \\ &= \mathbb{E}[\text{Cov}[\mathbf{f}_{\pi,m}(Z_{i|k}) + \tilde{\mathbf{g}}_{\pi,m}(Z_{i|k}), \\ &\quad \mathbf{f}_{\pi,n}(Z_{i|k}) + \tilde{\mathbf{g}}_{\pi,n}(Z_{i|k}) | X_{i|k}]] \\ &\quad + \text{Cov}[\mathbb{E}[\mathbf{f}_{\pi,m}(Z_{i|k}) + \tilde{\mathbf{g}}_{\pi,m}(Z_{i|k}) | X_{i|k}], \\ &\quad \mathbb{E}[\mathbf{f}_{\pi,n}(Z_{i|k}) + \tilde{\mathbf{g}}_{\pi,n}(Z_{i|k}) | X_{i|k}]] \\ &= \mathbb{E}[\text{Cov}[\tilde{\mathbf{g}}_{\pi,m}(Z_{i|k}), \tilde{\mathbf{g}}_{\pi,n}(Z_{i|k}) | X_{i|k}]] \\ &\quad + \text{Cov}[\mathbf{f}_{\pi,m}(Z_{i|k}) + \mu_i(Z_{i|k}), \mathbf{f}_{\pi,n}(Z_{i|k}) + \mu_j(Z_{i|k})], \quad (20) \end{aligned}$$

where $\mathbf{f}_{\pi,m}, \mathbf{f}_{\pi,n}, \tilde{\mathbf{g}}_{\pi,m}$, and $\tilde{\mathbf{g}}_{\pi,n}$ denote the i th and j th elements of $\mathbf{f}_\pi$ and $\tilde{\mathbf{g}}_\pi$, respectively. Since the elements of $\mathbf{g}_\pi$ are independent, the result in (20) can be simplified with

$$\mathbb{E}[\text{Cov}[\tilde{\mathbf{g}}_{\pi,m}(Z_{i|k}), \tilde{\mathbf{g}}_{\pi,n}(Z_{i|k}) | X_{i|k}]] = \begin{cases} 0, & \text{if } m \neq n \\ \mathbb{E}[\sigma_m^2(Z_{i|k})], & \text{if } m = n. \end{cases} \quad (21)$$

Inserting (21) in (20) and arranging the covariances as matrix results in

$$\text{Cov}[X_{i+1|k}] = \text{Cov}[\mathbf{f}_\pi(Z_{i|k}) + \mu(Z_{i|k})] + \mathbb{E}[\Sigma(Z_{i|k})]. \quad \square$$

The following example shows the propagation of the mean and the variance with (18) and (19) for a one-dimensional system.

Example 1. Consider the nonlinear system

$$x_{k+1} = \underbrace{x_k + 0.6 \sin(x_k)}_{f_\pi(x_k)} + \underbrace{\cos(x_k)}_{g_\pi(x_k)} \quad (22)$$

with state $x \in [-\pi, \pi]$, known system function $f_\pi(x_k)$, and unknown system function $g_\pi(x_k)$. The unknown part is learned by a GP $\tilde{g}_\pi \sim \mathcal{GP}(0, k)$ with squared exponential kernel

$$k(x, x') = C \exp(-\beta(x - x')^2) \quad (23)$$

with parameters $C = 1$ and $\beta = \frac{1}{2}$. Three noise-free data points at $x = -0.5$, $x = 0.5$, and $x = 1.5$ are available which fulfill Assumption 3. Figure 1a shows the unknown system function g_π as well as the mean and the 95% confidence interval of the GP. It demonstrates that the uncertainty about the value of g_π increases significantly if $\tilde{g}_\pi$ is evaluated at a point that is far away from a data point. This uncertainty not only affects the next predicted state but also has an impact on predicted states further in the future. Figure 1b shows the state trajectory according to (22) starting at $x_k = -0.9$ and the corresponding mean and standard deviation of the predicted state $X_{i|k}$ which are calculated by (18) and (19). Even though the standard deviation increases over time in this example, this is not always the case. If the predicted state converges to an equilibrium, for example, the standard deviation of the predicted state can also decrease.

3.2 | Error Bound of GP Predictions

Guarantees regarding recursive feasibility and stability of a controller require that the predicted states may not deviate arbitrarily from future states. Thus, the aim of this subsection is to find an upper bound on the difference between the expected value of the predicted states and the true state in the future. To this end, an upper bound for the difference between the mean of the GP posterior and the value of the unknown function g_π is required, which is stated in the following lemma.

Lemma 3. Let Assumptions 1 and 3 hold. Then, the expected error of the posterior mean is bounded by

$$\mathbb{E}[\|\mu(Z_{i|k}) - g_\pi(Z_{i|k})\|] \leq b_{\max} \sqrt{\mathbb{E}[\text{tr}(\Sigma(Z_{i|k}))]} \quad (24)$$

with $b_{\max}$ from (7) and $i \geq 0$.

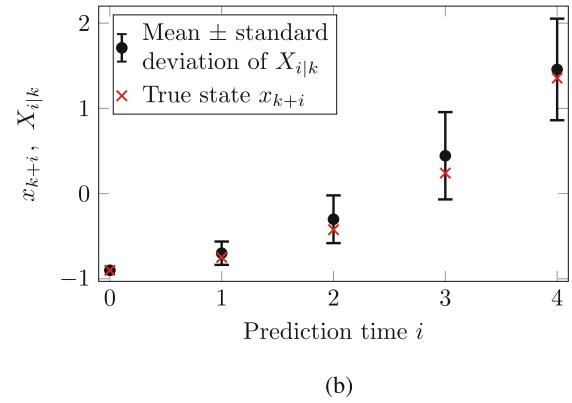
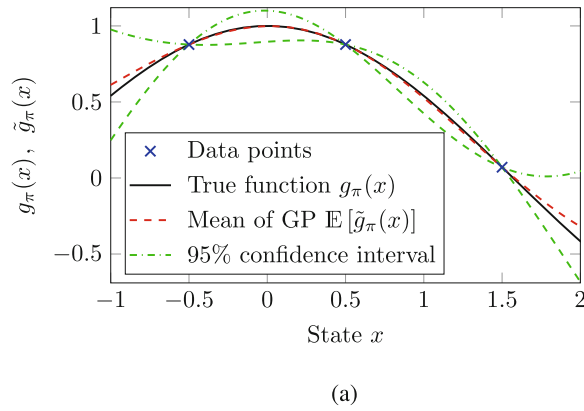


FIGURE 1 | Approximation of the unknown system function in (22) by GP with three data points (a) unknown system function and approximation by a GP, (b) true state trajectory and predicted state.

Proof. As shown in [12, Corollary 3.11], the error of the posterior mean of each element of $\mathbf{g}_\pi$ is bounded by

$$(\mu_j(\mathbf{z}) - g_{\pi,j}(\mathbf{z}))^2 \leq b_j^2 \sigma_j^2(\mathbf{z}) \quad (25)$$

for all $j \in \llbracket 1, N_x \rrbracket$ and a deterministic $\mathbf{z} \in \mathbb{Z}$, where $\mu_j(\mathbf{z})$ and $\sigma_j^2(\mathbf{z})$ are given by (14) and (15). For a random variable $Z_{i|k}$ follows by taking the expectation on both sides of (25) and summing all elements

$$\begin{aligned} \sum_{j=1}^{N_x} \mathbb{E}[(\mu_j(Z_{i|k}) - g_{\pi,j}(Z_{i|k}))^2] &\leq \sum_{j=1}^{N_x} b_j^2 \mathbb{E}[\sigma_j^2(Z_{i|k})] \\ \mathbb{E}\left[\sum_{j=1}^{N_x} (\mu_j(Z_{i|k}) - g_{\pi,j}(Z_{i|k}))^2\right] &\leq b_{\max}^2 \mathbb{E}\left[\sum_{j=1}^{N_x} \sigma_j^2(Z_{i|k})\right] \\ \mathbb{E}\left[\|\boldsymbol{\mu}(Z_{i|k}) - \mathbf{g}_\pi(Z_{i|k})\|^2\right] &\leq b_{\max}^2 \mathbb{E}[\text{tr}(\boldsymbol{\Sigma}(Z_{i|k}))], \end{aligned} \quad (26)$$

where the definition of the Euclidean norm is used in the last line. Applying Jensen's inequality results in

$$\mathbb{E}\left[\|\boldsymbol{\mu}(Z_{i|k}) - \mathbf{g}_\pi(Z_{i|k})\|^2\right] \leq \mathbb{E}\left[\|\boldsymbol{\mu}(Z_{i|k}) - \mathbf{g}_\pi(Z_{i|k})\|^2\right],$$

which can be inserted in (26) and leads to (24). $\square$

Next, an upper bound for $\|\mathbf{x}_{k+i} - \mathbb{E}[X_{i|k}]\|$ is derived. Therefore, the variables

$$\delta_{i|k} := \|\mathbf{x}_{k+i} - \mathbb{E}[X_{i|k}]\| \quad (27)$$

$$c_{i|k} := \sqrt{\text{tr}(\text{Cov}[X_{i|k}])} \quad (28)$$

$$\sigma_{\max} := \sqrt{\sum_{j=1}^{N_x} \|k_j\|_\infty} \quad (29)$$

are introduced to simplify the notation. Note that the kernels k_j in (29) are assumed to be bounded in Assumption 1 and hence $\sigma_{\max}$ is finite. Since $\sigma_j^2(\mathbf{z}) \leq k_j(\mathbf{z}, \mathbf{z})$ follows directly from (15),

$$\mathbb{E}[\text{tr}(\boldsymbol{\Sigma}(Z))] \leq \sigma_{\max}^2 \quad (30)$$

holds for all random variables $Z \in \mathbb{Z}$. This is used in the following lemma.

Lemma 4. Let Assumptions 1–4 hold. Then,

$$c_{i|k} \leq \hat{c}_i \quad (31)$$

for all $i \geq 0$ with

$$\hat{c}_i := \begin{cases} 0, & i = 0 \\ \sigma_{\max} \sqrt{\sum_{j=0}^{i-1} (L_f + L_\mu)^{2j}}, & i > 0. \end{cases} \quad (32)$$

Proof. In order to prove Lemma 4, we aim to find a recursive law that bounds $c_{i|k}$ depending on $c_{i-1|k}$ for $i > 0$. To this end, we use a standard approach from the algebra of random variables, see for example [30]. Assume that we have two iid random variables $X_{i|k}$ and $Y_{i|k}$. Then, it holds for the j th element with $j \in \llbracket 1, N_x \rrbracket$ that

$$\begin{aligned} \text{Var}[X_{i|k,j}] &= \frac{1}{2} (\text{Var}[X_{i|k,j}] + \text{Var}[Y_{i|k,j}]) \\ &= \frac{1}{2} (\text{Var}[X_{i|k,j}] + \text{Var}[-Y_{i|k,j}]) \\ &= \frac{1}{2} \text{Var}[X_{i|k,j} - Y_{i|k,j}] \end{aligned} \quad (33)$$

because $X_{i|k,j}$ and $Y_{i|k,j}$ are iid. The definition of $c_{i|k}$ in (28) is based on the trace of the covariance matrix, for which holds that

$$\text{tr}(\text{Cov}[X_{i|k}]) = \sum_{j=1}^{N_x} \text{Var}[X_{i|k,j}] = \frac{1}{2} \sum_{j=1}^{N_x} \text{Var}[X_{i|k,j} - Y_{i|k,j}],$$

where (33) has been used in the last step. Applying the definition of the variance results in

$$\text{tr}(\text{Cov}[X_{i|k}]) = \frac{1}{2} \sum_{j=1}^{N_x} (\mathbb{E}[(X_{i|k,j} - Y_{i|k,j})^2] - \mathbb{E}[X_{i|k,j} - Y_{i|k,j}]^2).$$

Since $X_{i|k,j}$ and $Y_{i|k,j}$ are iid, the expected value of the difference is zero and the remaining expectation can be shifted outside the sum which leads to

$$\begin{aligned}\text{tr}(\text{Cov}[X_{i|k}]) &= \frac{1}{2} \mathbb{E} \left[\sum_{j=1}^{N_s} (X_{i|k,j} - Y_{i|k,j})^2 \right] \\ &= \frac{1}{2} \mathbb{E} [\|X_{i|k} - Y_{i|k}\|^2].\end{aligned}\quad (34)$$

Next, we define $Z_{i|k} = (X_{i|k}, \mathbf{v})$ and $Z'_{i|k} = (X_{i|k}, \mathbf{v})$ which are iid. In the same way as before, it can be shown that

$$\begin{aligned}\text{tr}(\text{Cov}[\mathbf{f}_\pi(Z_{i|k}) + \boldsymbol{\mu}(Z_{i|k})]) \\ = \frac{1}{2} \mathbb{E} [\|\mathbf{f}_\pi(Z_{i|k}) + \boldsymbol{\mu}(Z_{i|k}) - \mathbf{f}_\pi(Z'_{i|k}) - \boldsymbol{\mu}(Z'_{i|k})\|^2].\end{aligned}\quad (35)$$

The function $\mathbf{f}_\pi$ is Lipschitz continuous with Lipschitz constant L_f and the function $\boldsymbol{\mu}$ is Lipschitz continuous with Lipschitz constant L_μ due to Assumptions 2 and 4, respectively. Thus, the sum of both is Lipschitz continuous with Lipschitz constant $L_f + L_\mu$ and (35) can be bounded by

$$\text{tr}(\text{Cov}[\mathbf{f}_\pi(Z_{i|k}) + \boldsymbol{\mu}(Z_{i|k})]) \leq \frac{1}{2} \mathbb{E} [(L_f + L_\mu)^2 \|X_{i|k} - Y_{i|k}\|^2]$$

and with (34), we have

$$\text{tr}(\text{Cov}[\mathbf{f}_\pi(Z_{i|k}) + \boldsymbol{\mu}(Z_{i|k})]) \leq (L_f + L_\mu)^2 \text{tr}(\text{Cov}[X_{i|k}]). \quad (36)$$

The combination of (28), (19), (30), and (36) results in

$$\begin{aligned}c_{i+1|k} &= \sqrt{\text{tr}(\text{Cov}[X_{i+1|k}])} \\ &= \sqrt{\text{tr}(\text{Cov}[\mathbf{f}_\pi(Z_{i|k}) + \boldsymbol{\mu}(Z_{i|k})] + \mathbb{E}[\boldsymbol{\Sigma}(Z_{i|k})])} \\ &\leq \sqrt{(L_f + L_\mu)^2 \text{tr}(\text{Cov}[X_{i|k}]) + \text{tr}(\mathbb{E}[\boldsymbol{\Sigma}(Z_{i|k})])} \\ &\leq \sqrt{(L_f + L_\mu)^2 c_{i|k}^2 + \sigma_{\max}^2}\end{aligned}$$

for $i \geq 0$. Starting from $c_{0|k} = 0$, the upper bound is equivalent to (32). $\square$

Next, an upper bound of the prediction error (27) is derived. For this purpose, the following lemma is necessary as an intermediate step.

Lemma 5. *The expected value of the norm of the predicted state error is bounded by*

$$\mathbb{E} [\|\mathbf{x}_{k+i} - X_{i|k}\|] \leq \delta_{i|k} + c_{i|k} \quad (37)$$

for all $i \geq 0$.

Proof. Expanding the left side of (37) results in

$$\mathbb{E} [\|\mathbf{x}_{k+i} - X_{i|k}\|] = \mathbb{E} [\|\mathbf{x}_{k+i} - \mathbb{E}[X_{i|k}] + \mathbb{E}[X_{i|k}] - X_{i|k}\|]$$

and the triangle inequality states that

$$\mathbb{E} [\|\mathbf{x}_{k+i} - X_{i|k}\|] \leq \mathbb{E} [\|\mathbf{x}_{k+i} - \mathbb{E}[X_{i|k}]\|] + \mathbb{E} [\|\mathbb{E}[X_{i|k}] - X_{i|k}\|], \quad (38)$$

where the right-hand side can be rewritten with

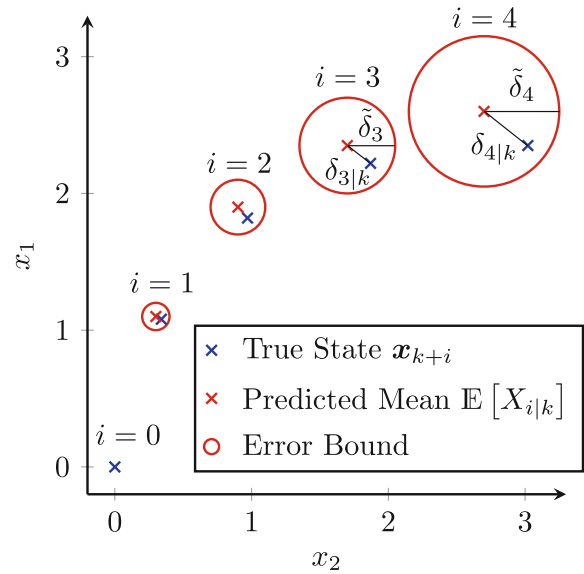


FIGURE 2 | Difference between the mean of the predicted state using the GP-prediction and the true state trajectory according to (4). The error bound $\tilde{\delta}_i$ with $i \geq 0$ is shown as a red circle around the predicted mean.

$$\mathbb{E} [\|\mathbb{E}[X_{i|k}] - X_{i|k}\|] = \mathbb{E} \left[\sqrt{\sum_{j=1}^{N_s} (X_{i|k,j} - \mathbb{E}[X_{i|k,j}])^2} \right].$$

Since the square root is a concave function, Jensen's inequality can be applied to (38) which results in

$$\begin{aligned}\mathbb{E} [\|\mathbf{x}_{k+i} - X_{i|k}\|] \\ \leq \|\mathbf{x}_{k+i} - \mathbb{E}[X_{i|k}]\| + \sqrt{\sum_{j=1}^{N_s} \mathbb{E} [(X_{i|k,j} - \mathbb{E}[X_{i|k,j}])^2]}.\end{aligned}$$

Using the definition of the variance

$$\text{Var}[X_{i|k,j}] = \mathbb{E} [(X_{i|k,j} - \mathbb{E}[X_{i|k,j}])^2], \quad j \in [1, N_s]$$

leads to (37) because the diagonal elements of $\text{Cov}[X_{i|k}]$ are the corresponding variances. $\square$

Figure 2 shows an example of the predicted expected value of the state and the true future state of a two-dimensional system. In addition, a circular error bound $\tilde{\delta}$ is shown, which is an upper limit of $\delta_{i|k}$ for all $i \geq 0$. The following proposition derives this upper bound based on Lemmas 4 and 5.

Proposition 1. *Let Assumptions 1–4 hold. Then, the norm of the error between the mean of the predicted state and the true state is bounded by*

$$\delta_{i+1|k} \leq \tilde{\delta}_{i+1} \quad (39)$$

for all $i \geq 0$ with

$$\begin{aligned}\tilde{\delta}_0 &= 0 \\ \tilde{\delta}_{i+1} &= (L_f + L_g)(\tilde{\delta}_i + \hat{c}_i) + b_{\max} \sigma_{\max}.\end{aligned}\quad (40)$$

Proof. It follows from (27) and Lemma 2 that

$$\delta_{i+1|k} = \left\| \mathbf{f}_\pi(\mathbf{z}_{k+i}) + \mathbf{g}_\pi(\mathbf{z}_{k+i}) - \mathbb{E}[\mathbf{f}_\pi(Z_{i|k}) + \boldsymbol{\mu}(Z_{i|k})] \right\|$$

and by applying the triangle inequality

$$\delta_{i+1|k} \leq \left\| \mathbf{f}_\pi(\mathbf{z}_{k+i}) - \mathbb{E}[\mathbf{f}_\pi(Z_{i|k})] \right\| + \left\| \mathbf{g}_\pi(\mathbf{z}_{k+i}) - \mathbb{E}[\boldsymbol{\mu}(Z_{i|k})] \right\|,$$

which is equivalent to

$$\delta_{i+1|k} \leq \left\| \mathbb{E}[\mathbf{f}_\pi(\mathbf{z}_{k+i}) - \mathbf{f}_\pi(Z_{i|k})] \right\| + \left\| \mathbb{E}[\mathbf{g}_\pi(\mathbf{z}_{k+i}) - \boldsymbol{\mu}(Z_{i|k})] \right\|.$$

Since the Euclidean norm is a convex function, Jensen's inequality can be applied, which results in

$$\delta_{i+1|k} \leq \mathbb{E} \left[\left\| \mathbf{f}_\pi(\mathbf{z}_{k+i}) - \mathbf{f}_\pi(Z_{i|k}) \right\| \right] + \mathbb{E} \left[\left\| \mathbf{g}_\pi(\mathbf{z}_{k+i}) - \boldsymbol{\mu}(Z_{i|k}) \right\| \right],$$

where the right-hand side can be expanded to

$$\begin{aligned} \delta_{i+1|k} &\leq \mathbb{E} \left[\left\| \mathbf{f}_\pi(\mathbf{z}_{k+i}) - \mathbf{f}_\pi(Z_{i|k}) \right\| \right] \\ &\quad + \mathbb{E} \left[\left\| \mathbf{g}_\pi(\mathbf{z}_{k+i}) - \boldsymbol{\mu}(Z_{i|k}) + \mathbf{g}_\pi(Z_{i|k}) - \mathbf{g}_\pi(Z_{i|k}) \right\| \right]. \end{aligned}$$

Applying the triangle inequality leads to

$$\begin{aligned} \delta_{i+1|k} &\leq \mathbb{E} \left[\left\| \mathbf{f}_\pi(\mathbf{z}_{k+i}) - \mathbf{f}_\pi(Z_{i|k}) \right\| \right] + \mathbb{E} \left[\left\| \mathbf{g}_\pi(\mathbf{z}_{k+i}) - \mathbf{g}_\pi(Z_{i|k}) \right\| \right] \\ &\quad + \mathbb{E} \left[\left\| \mathbf{g}_\pi(Z_{i|k}) - \boldsymbol{\mu}(Z_{i|k}) \right\| \right] \end{aligned}$$

and with Assumption 2, Lemmas 1, and 3, we have

$$\delta_{i+1|k} \leq (L_f + L_g) \mathbb{E} \left[\left\| \mathbf{x}_{k+i} - X_{i|k} \right\| \right] + b_{\max} \sqrt{\mathbb{E}[\text{tr}(\boldsymbol{\Sigma}(Z_{i|k}))]}.$$

Applying Lemma 5 results in

$$\delta_{i+1|k} \leq (L_f + L_g)(\delta_{i|k} + c_{i|k}) + b_{\max} \sqrt{\mathbb{E}[\text{tr}(\boldsymbol{\Sigma}(Z_{i|k}))]} \quad (41)$$

and using (30) and Lemma 4, it follows from (41) that

$$\delta_{i+1|k} \leq (L_f + L_g)(\tilde{\delta}_i + \hat{c}_i) + b_{\max} \sigma_{\max}. \quad \square$$

4 | Stochastic MPC

The GP model $\tilde{\mathbf{g}}_\pi$ is used in the design of a stochastic model predictive controller. Since the system (4) is partially unknown, the controller uses the prediction model

$$X_{i+1|k} = \mathbf{f}_\pi(Z_{i|k}) + \boldsymbol{\mu}(Z_{i|k}) + W_{i|k}(Z_{i|k}), \quad i \in \mathbb{I}_{[0, N-1]}, \quad (42)$$

where $N > 0$ is the prediction horizon and $W_{i|k}$ is a zero-mean Gaussian distributed random variable with covariance matrix $\text{Cov}[W_{i|k}(Z_{i|k})] = \boldsymbol{\Sigma}(Z_{i|k})$. Note that the disturbance $W_{i|k}$ is not assumed to be bounded. Nevertheless, the control method proposed in Section 4.1 guarantees recursive feasibility and stability as shown in Sections 4.2 and 4.3.

4.1 | SMPC Formulation

Let

$$\bar{\mathbf{v}}_k = (\mathbf{v}_{0|k}, \mathbf{v}_{1|k}, \dots, \mathbf{v}_{N-1|k}) \quad (43)$$

be a sequence of control policy parameters for the time interval from k to $k + N - 1$ and let

$$\bar{W}_k = (W_{0|k}, W_{1|k}, \dots, W_{N-1|k}) \quad (44)$$

be the corresponding sequence of disturbances, where the dependencies are neglected in the notation for the sake of simplicity. The cost function depending on $\bar{\mathbf{v}}_k$ and $\bar{W}_k$ is defined by

$$J_N(\mathbf{x}_k, \bar{\mathbf{v}}_k, \bar{W}_k) = \sum_{i=0}^{N-1} \ell(X_{i|k}, \pi(X_{i|k}, \mathbf{v}_{i|k})) + V_f(X_{N|k}) \quad (45)$$

with the Borel-measurable running cost function $\ell : \mathbb{X} \times \mathbb{U} \rightarrow \mathbb{R}_{\geq 0}$ and the Borel-measurable terminal cost $V_f : \mathbb{X} \rightarrow \mathbb{R}_{\geq 0}$. The result of (45) is a random variable because the predicted states are random variables. Thus, it is common in the context of SMPC to use the expected value

$$V_N(\mathbf{x}_k, \bar{\mathbf{v}}_k) = \mathbb{E} \left[J_N(\mathbf{x}_k, \bar{\mathbf{v}}_k, \bar{W}_k) \right] \quad (46)$$

as the objective function of an OCP.

The controller must satisfy constraints

$$(\mathbf{x}_{k+i}, \mathbf{v}_{i|k}) \in \mathcal{Z} \quad \forall i \in \mathbb{I}_{[0, N-1]}$$

with $\mathcal{Z}$ according to (5). Furthermore, a terminal region must be reached at the prediction horizon by the terminal constraint

$$\mathbf{x}_{k+N} \in \mathbb{X}_f, \quad (47)$$

which is defined similar to $\mathcal{Z}$ by a set of inequalities

$$\mathbb{X}_f = \{\mathbf{x} \in \mathbb{X} \mid q_j(\mathbf{x}) \leq 0 \quad \forall j \in \mathbb{I}_{[1, N_q]}\} \quad (48)$$

with Borel-measurable terminal constraint functions $q_j : \mathbb{X} \rightarrow \mathbb{R}$, where N_q is the number of terminal constraint functions. These functions must fulfill the following assumption.

Assumption 5. The functions q_j with $j \in \mathbb{I}_{[1, N_q]}$ are Lipschitz continuous with respect to $\mathbf{x}$ such that

$$\left| q_j(\mathbf{x}) - q_j(\mathbf{x}') \right| \leq L_{q,j} \|\mathbf{x} - \mathbf{x}'\| \quad (49)$$

holds for all $\mathbf{x}, \mathbf{x}' \in X_f$, where $L_{q,j}$ is the Lipschitz constant of q_j .

Proposition 1 states a valid upper bound for the prediction error (27). However, the following modification to the upper bound is required in order to guarantee recursive feasibility of the control in Section 4.2.

Corollary 1. Let Assumptions 1–4 hold. Then, the norm of the error between the mean of the predicted state and the true state is bounded by

$$\delta_{i+1|k} \leq \hat{\delta}_{i+1} \quad (50)$$

for all $i \geq 0$ with

$$\begin{aligned}\hat{\delta}_0 &= 0 \\ \hat{\delta}_{i+1} &= (L_f + \max(L_g, L_\mu))(\hat{\delta}_i + \hat{c}_i) + b_{\max}\sigma_{\max} + \hat{\delta}_i,\end{aligned}\quad (51)$$

where L_μ is the Lipschitz constant of μ from Assumption 4.

Proof. Comparing the definitions of $\tilde{\delta}_{i+1}$ in (40) and $\hat{\delta}_{i+1}$ in (51) results in $\tilde{\delta}_{i+1} \leq \hat{\delta}_{i+1}$ for all $i \geq 0$. Using (39), we have

$$\delta_{i+1|k} \leq \tilde{\delta}_{i+1} \leq \hat{\delta}_{i+1} \quad (52)$$

for all $i \geq 0$. $\square$

The stochastic model predictive controller must guarantee the satisfaction of the constraints (5) and (47) even though the prediction model (42) may differ from the true system dynamics (4). To this end, the OCP which is solved by the controller considers tightened constraints defined by the sets

$$\begin{aligned}\tilde{\mathcal{Z}}_i &:= \{(\mathbf{x}, \mathbf{v}) \in \mathbb{X} \times \mathbb{V} | h_j(\mathbf{x}, \mathbf{v}) \\ &\leq -L_{h,j} \hat{\delta}_i \quad \forall j \in \mathbb{I}_{[0, N_h-1]}\} \subseteq \mathcal{Z}\end{aligned}\quad (53)$$

$$\begin{aligned}\tilde{\mathbb{X}}_f &:= \left\{ \mathbf{x} \in \mathbb{X} | q_j(\mathbf{x}) \leq -L_{q,j} \hat{\delta}_N \quad \forall j \in \mathbb{I}_{[0, N_q-1]} \right\} \\ &\cap \{ \mathbf{x} \in \mathbb{X} | (\mathbf{x}, \hat{\mathbf{v}}) \in \tilde{\mathcal{Z}}_N \},\end{aligned}\quad (54)$$

where $\hat{\mathbf{v}}$ is the parameterization that belongs to a control law $\kappa_f(\mathbf{x})$ for the terminal region which fulfills the following assumption. It is in most parts identical to the assumption for SMPC in [20].

Assumption 6. There exists a terminal region $\mathbb{X}_f \subseteq \mathbb{X}$ according to (48), a continuous control law $\mathbf{u} = \kappa_f(\mathbf{x}) \in \mathbb{U}$, and a constant $\epsilon \geq 0$ such that

$$V_f(\mathbf{f}(\mathbf{x}, \kappa_f(\mathbf{x})) + \mu(\mathbf{x}, \kappa_f(\mathbf{x}))) \leq V_f(\mathbf{x}) - \ell(\mathbf{x}, \kappa_f(\mathbf{x})) \quad (55)$$

$$\mathbf{f}(\mathbf{x}, \kappa_f(\mathbf{x})) + \mathbf{g}(\mathbf{x}, \kappa_f(\mathbf{x})) \in \tilde{\mathbb{X}}_f \quad (56)$$

$$\begin{aligned}\mathbb{E}[V_f(\mathbf{f}(\mathbf{x}, \kappa_f(\mathbf{x})) + \mu(\mathbf{x}, \kappa_f(\mathbf{x})) + W(\mathbf{x}, \kappa_f(\mathbf{x}))) \\ \leq V_f(\mathbf{x}) - \ell(\mathbf{x}, \kappa_f(\mathbf{x})) + \epsilon\end{aligned}\quad (57)$$

for all $\mathbf{x} \in \mathbb{X}_f$ and $W(\mathbf{x}, \kappa_f(\mathbf{x})) \sim \mathcal{N}(\mathbf{0}, \Sigma(\mathbf{x}, \kappa_f(\mathbf{x})))$ and there exists a parameterization $\hat{\mathbf{v}} \in \mathbb{V}$ such that $\pi(\mathbf{x}, \hat{\mathbf{v}}) = \kappa_f(\mathbf{x})$ and $(\mathbf{x}, \hat{\mathbf{v}}) \in \mathcal{Z}$ for all $\mathbf{x} \in \mathbb{X}_f$.

The stochastic OCP $\mathbb{P}_N(\mathbf{x}_k)$ that is solved by the model predictive controller in each time step is defined as

$$\min_{\bar{\mathbf{v}}_k} V_N(\mathbf{x}_k, \bar{\mathbf{v}}_k) \quad (58a)$$

$$s.t. \quad X_{i+1|k} = \mathbf{f}_\pi(Z_{i|k}) + \mu(Z_{i|k}) + W_{i|k}(Z_{i|k}) \quad (58b)$$

$$(\mathbb{E}[X_{i|k}], \mathbf{v}_{i|k}) \in \tilde{\mathcal{Z}}_i \quad (58c)$$

$$\mathbb{E}[X_{N|k}] \in \tilde{\mathbb{X}}_f \quad (58d)$$

$$X_{0|k} = \mathbf{x}_k, \quad i \in \mathbb{I}_{[0, N-1]} \quad (58e)$$

with tightened constraint sets according to (53) and (54). The set of initial states for which $\mathbb{P}_N$ is feasible is denoted by $\mathcal{X}_N$. The sequence of control policy parameters that solves $\mathbb{P}_N(\mathbf{x}_k)$ is denoted by

$$\bar{\mathbf{v}}_k^* = \left(\mathbf{v}_{0|k}^*, \mathbf{v}_{1|k}^*, \dots, \mathbf{v}_{N-1|k}^* \right), \quad (59)$$

the corresponding cost is denoted by

$$V_N^*(\mathbf{x}_k) := V_N(\mathbf{x}_k, \bar{\mathbf{v}}_k^*), \quad (60)$$

and the control law that is generated by the stochastic model predictive controller is defined as

$$\kappa_N(\mathbf{x}_k) := \pi(\mathbf{x}_k, \mathbf{v}_{0|k}^*) \quad (61)$$

for $\mathbf{x}_k \in \mathcal{X}_N$.

4.2 | Constraint Satisfaction and Recursive Feasibility

The OCP (58) does not consider the original constraints (5) and (47) but tightened constraints (58c) and (58d) in order to account for the uncertainty. The following theorem shows that the constraint (5) and the terminal constraint (47) are fulfilled by the state trajectory from time step k to $k+N$ if a control policy sequence is applied to the system which is a feasible solution of the OCP (58).

Theorem 1. Let Assumptions 1–4 hold and let $\bar{\mathbf{v}}_k$ be a feasible solution of $\mathbb{P}_N(\mathbf{x}_k)$ with $\mathbf{x}_k \in \mathcal{X}_N$. Then, by applying $\mathbf{v}_{k+i} = \mathbf{v}_{i|k}$ with $i \in \mathbb{I}_{[0, N-1]}$ to the system that is in state $\mathbf{x}_k$, it holds for the future states

$$(\mathbf{x}_{k+i}, \mathbf{v}_{k+i}) \in \mathcal{Z} \quad (62)$$

$$\mathbf{x}_{k+N} \in \mathbb{X}_f \quad (63)$$

if

$$(\mathbb{E}[X_{i|k}], \mathbf{v}_{i|k}) \in \tilde{\mathcal{Z}}_i \quad (64)$$

$$\mathbb{E}[X_{N|k}] \in \tilde{\mathbb{X}}_f \quad (65)$$

with $\tilde{\mathcal{Z}}_i$ and $\tilde{\mathbb{X}}_f$ according to (53) and (54).

Proof. Assumption 2 and Corollary 1 give the upper bound

$$\begin{aligned}h_j(\mathbf{x}_{k+i}, \mathbf{v}_{k+i}) &\leq h_j(\mathbb{E}[X_{i|k}], \mathbf{v}_{i|k}) + L_{h,j} \delta_{i|k} \\ &\leq h_j(\mathbb{E}[X_{i|k}], \mathbf{v}_{i|k}) + L_{h,j} \hat{\delta}_i\end{aligned}\quad (66)$$

for all $j \in \mathbb{I}_{[1, N_h]}$ and for $\mathbf{v}_{k+i} = \mathbf{v}_{i|k}$, where $\hat{\delta}_i$ can be calculated recursively with (51) starting from $\hat{\delta}_0 = 0$. Thus,

$$h_j(\mathbf{x}_{k+i}, \mathbf{v}_{k+i}) \leq 0 \quad \forall j \in \mathbb{I}_{[1, N_h]} \quad (67)$$

if

$$h_j(\mathbb{E}[X_{i|k}], \mathbf{v}_{i|k}) < -L_{h,j} \hat{\delta}_i \quad \forall j \in \mathbb{I}_{[1, N_h]}. \quad (68)$$

From (68) and the definitions of $\tilde{z}_i$ in (53) and $\mathcal{Z}$ in (5) follows directly that (62) holds if (64) is fulfilled.

In the same way, it can be proven that (63) follows from (65) using the Lipschitz constants in Assumption 5. $\square$

In a next step, it is shown that the OCP (58) is recursively feasible. Therefore, the sequence of control policy parameters

$$\begin{aligned}\tilde{\mathbf{v}}_{k+1} &= (\tilde{\mathbf{v}}_{0|k+1}, \tilde{\mathbf{v}}_{1|k+1}, \dots, \tilde{\mathbf{v}}_{N-1|k+1}) \\ &:= (\mathbf{v}_{1|k}^*, \mathbf{v}_{2|k}^*, \dots, \mathbf{v}_{N-1|k}^*, \hat{\mathbf{v}})\end{aligned}\quad (69)$$

is defined which is a shifted version of $\bar{\mathbf{v}}_k^*$ with $\hat{\mathbf{v}}$ added at the end of the sequence. The following two lemmas are necessary as intermediate steps for the subsequent theorem.

Lemma 6. *Let Assumptions 1–4 hold. Then it holds that*

$$\mathbb{E}[\|\mathbf{X}_{i-1|k+1} - \mathbf{X}_{i|k}\|] \leq (L_f + L_\mu)^{i-1} \sigma_{\max}(b_{\max} + 1) \quad (70)$$

for $i \in \mathbb{I}_{[1,N]}$ if $\bar{\mathbf{v}}_k^*$ is used to generate the predictions at time step k and $\tilde{\mathbf{v}}_{k+1}$ is used to generate the predictions at time step $k+1$.

Proof. From Lemma 5, we have

$$\mathbb{E}[\|\mathbf{X}_{0|k+1} - \mathbf{X}_{1|k}\|] \leq \delta_{1|k} + c_{1|k} \quad (71)$$

and from Lemma 4 and Corollary 1

$$\mathbb{E}[\|\mathbf{X}_{0|k+1} - \mathbf{X}_{1|k}\|] \leq \hat{\delta}_1 + \hat{c}_1 = \sigma_{\max}(b_{\max} + 1). \quad (72)$$

For $i \in \mathbb{I}_{[2,N]}$, the expected value of the norm of the difference between the predicted states can be written as

$$\begin{aligned}\mathbb{E}[\|\mathbf{X}_{i-1|k+1} - \mathbf{X}_{i|k}\|] \\ = \mathbb{E}[\|\mathbf{f}_\pi(\mathbf{Z}_{i-2|k+1}) + \boldsymbol{\mu}(\mathbf{Z}_{i-2|k+1}) - \mathbf{f}_\pi(\mathbf{Z}_{i-1|k}) - \boldsymbol{\mu}(\mathbf{Z}_{i-1|k})\|],\end{aligned}$$

for which the upper bound

$$\begin{aligned}\mathbb{E}[\|\mathbf{X}_{i-1|k+1} - \mathbf{X}_{i|k}\|] &\leq \mathbb{E}[\|\mathbf{f}_\pi(\mathbf{Z}_{i-2|k+1}) - \mathbf{f}_\pi(\mathbf{Z}_{i-1|k})\|] \\ &\quad + \mathbb{E}[\|\boldsymbol{\mu}(\mathbf{Z}_{i-2|k+1}) - \boldsymbol{\mu}(\mathbf{Z}_{i-1|k})\|]\end{aligned}$$

is given by the triangle inequality. Using the Lipschitz constants in Assumptions 2 and 4, the upper bound can be stated as the recursive formula

$$\mathbb{E}[\|\mathbf{X}_{i-1|k+1} - \mathbf{X}_{i|k}\|] \leq (L_f + L_\mu) \mathbb{E}[\|\mathbf{X}_{i-2|k+1} - \mathbf{X}_{i-1|k}\|]$$

for all $i \in \mathbb{I}_{[2,N]}$. Starting from (72) for $i=1$, this results in (70). $\square$

Lemma 7. *Let Assumptions 1–4 hold. Then it holds that*

$$\|\mathbb{E}[\mathbf{X}_{i|k+1}] - \mathbb{E}[\mathbf{X}_{i+1|k}]\| \leq \hat{\delta}_{i+1} - \hat{\delta}_i \quad \forall i \in \mathbb{I}_{[0,N-1]} \quad (73)$$

if $\bar{\mathbf{v}}_k^*$ is used to generate the predictions at time step k and $\tilde{\mathbf{v}}_{k+1}$ is used to generate the predictions at time step $k+1$.

Proof. The upper bound for $i=0$ is given by

$$\begin{aligned}\|\mathbb{E}[\mathbf{X}_{0|k+1}] - \mathbb{E}[\mathbf{X}_{1|k}]\| &= \|\mathbf{x}_{k+1} - \mathbb{E}[\mathbf{X}_{1|k}]\| \\ &= \delta_{1|k} \leq \hat{\delta}_1 \\ &= b_{\max} \sigma_{\max}\end{aligned}\quad (74)$$

from Corollary 1. Thus, (73) is fulfilled for $i=0$ because $\hat{\delta}_0 = 0$. The upper bound for $i>0$ can be found using the prediction model (42) and the triangle inequality that lead to

$$\begin{aligned}\|\mathbb{E}[\mathbf{X}_{i|k+1}] - \mathbb{E}[\mathbf{X}_{i+1|k}]\| &= \|\mathbb{E}[\mathbf{f}_\pi(\mathbf{Z}_{i-1|k+1}) + \boldsymbol{\mu}(\mathbf{Z}_{i-1|k+1})] \\ &\quad - \mathbb{E}[\mathbf{f}_\pi(\mathbf{Z}_{i|k}) + \boldsymbol{\mu}(\mathbf{Z}_{i|k})]\| \\ &\leq \|\mathbb{E}[\mathbf{f}_\pi(\mathbf{Z}_{i-1|k+1}) - \mathbf{f}_\pi(\mathbf{Z}_{i|k})]\| \\ &\quad + \|\mathbb{E}[\boldsymbol{\mu}(\mathbf{Z}_{i-1|k+1}) - \boldsymbol{\mu}(\mathbf{Z}_{i|k})]\|.\end{aligned}$$

Applying Jensen's inequality results in

$$\begin{aligned}\|\mathbb{E}[\mathbf{X}_{i|k+1}] - \mathbb{E}[\mathbf{X}_{i+1|k}]\| &\leq \mathbb{E}[\|\mathbf{f}_\pi(\mathbf{Z}_{i-1|k+1}) - \mathbf{f}_\pi(\mathbf{Z}_{i|k})\|] \\ &\quad + \mathbb{E}[\|\boldsymbol{\mu}(\mathbf{Z}_{i-1|k+1}) - \boldsymbol{\mu}(\mathbf{Z}_{i|k})\|]\end{aligned}$$

and with the Lipschitz continuity of $\mathbf{f}_\pi$ in Assumption 2 and the Lipschitz continuity of $\boldsymbol{\mu}$ in Assumption 4 it follows

$$\|\mathbb{E}[\mathbf{X}_{i|k+1}] - \mathbb{E}[\mathbf{X}_{i+1|k}]\| \leq (L_f + L_\mu) \mathbb{E}[\|\mathbf{X}_{i-1|k+1} - \mathbf{X}_{i|k}\|]$$

because $\tilde{\mathbf{v}}_{i-1|k+1} = \bar{\mathbf{v}}_{i|k}$ for all $i \in \mathbb{I}_{[1,N-1]}$. Applying Lemma 6 results in

$$\|\mathbb{E}[\mathbf{X}_{i|k+1}] - \mathbb{E}[\mathbf{X}_{i+1|k}]\| \leq (L_f + L_\mu)^i \sigma_{\max}(b_{\max} + 1). \quad (75)$$

Furthermore, it is known from Corollary 1 that

$$\begin{aligned}\hat{\delta}_{i+1} &\geq (L_f + L_\mu)(\hat{\delta}_i + \hat{c}_i) + \hat{\delta}_i \geq (L_f + L_\mu)^i \sigma_{\max}(b_{\max} + 1) + \hat{\delta}_i \\ &\quad \forall i \in \mathbb{I}_{[1,N-1]},\end{aligned}$$

from which

$$\hat{\delta}_{i+1} - \hat{\delta}_i \geq (L_f + L_\mu)^i \sigma_{\max}(b_{\max} + 1) \quad \forall i \in \mathbb{I}_{[1,N-1]} \quad (76)$$

follows directly. Combining (75) and (76) results in (73). $\square$

Theorem 2. *Let Assumptions 1–6, hold. Then, it holds for all $\mathbf{x}_k \in \mathcal{X}_N$ that $\mathbf{x}_{k+1} \in \mathcal{X}_N$ if $\mathbf{u}_k = \boldsymbol{\kappa}_N(\mathbf{x}_k)$. Thus, the stochastic OCP (58) is recursively feasible.*

Proof. Due to the definition of $\tilde{\mathbf{v}}_{k+1}$ in (69), we have

$$\begin{aligned}&\left| h_{\pi,j}(\mathbb{E}[\mathbf{X}_{i|k+1}], \tilde{\mathbf{v}}_{i|k+1}) - h_{\pi,j}(\mathbb{E}[\mathbf{X}_{i+1|k}], \mathbf{v}_{i+1|k}^*) \right| \\ &= \left| h_{\pi,j}(\mathbb{E}[\mathbf{X}_{i|k+1}], \tilde{\mathbf{v}}_{i|k+1}) - h_{\pi,j}(\mathbb{E}[\mathbf{X}_{i+1|k}], \tilde{\mathbf{v}}_{i|k+1}) \right|\end{aligned}$$

because $\tilde{\mathbf{v}}_{i|k+1} = \mathbf{v}_{i+1|k}^*$. From (11) and Assumption 6 follows for all $i \in \mathbb{I}_{[0,N-2]}$

$$\begin{aligned} & \left| h_{\pi,j}(\mathbb{E}[X_{i|k+1}], \tilde{\mathbf{v}}_{i|k+1}) - h_{\pi,j}(\mathbb{E}[X_{i+1|k}], \tilde{\mathbf{v}}_{i|k+1}) \right| \\ & \leq L_{h,j} \left\| \mathbb{E}[X_{i|k+1}] - \mathbb{E}[X_{i+1|k}] \right\| \quad \forall j \in \mathbb{I}_{[1, N_h]} \end{aligned}$$

and using Lemma 7, we have

$$h_{\pi,j}(\mathbb{E}[X_{i|k+1}], \tilde{\mathbf{v}}_{i|k+1}) \leq h_{\pi,j}(\mathbb{E}[X_{i+1|k}], \tilde{\mathbf{v}}_{i|k+1}) + L_{h,j}(\hat{\delta}_{i+1} - \hat{\delta}_i) \quad (77)$$

for all $i \in \mathbb{I}_{[0, N-2]}$ and $j \in \mathbb{I}_{[1, N_h]}$. Assuming that $\mathbf{x}_k \in \mathcal{X}_N$, the constraints (58c) are satisfied for all $i \in \mathbb{I}_{[0, N-2]}$ in time step k and an upper bound for the inequality constraints $h_{\pi,j}$ is known by (53). Thus, it follows from (77) that

$$h_{\pi,j}(\mathbb{E}[X_{i|k+1}], \tilde{\mathbf{v}}_{i|k+1}) \leq L_{h,j}(\hat{\delta}_{i+1} - \hat{\delta}_i - \hat{\delta}_{i+1}) = -L_{h,j}\hat{\delta}_i, \quad (78)$$

from which can be concluded that $(\mathbb{E}[X_{i|k+1}], \tilde{\mathbf{v}}_{i|k+1}) \in \tilde{\mathcal{Z}}_i$. The case $i = N - 1$ must be considered separately because (58c) is not required for $(\mathbb{E}[X_{N|k}], \mathbf{v}_{N|k})$. However, the definition of (54) ensures that $(\mathbb{E}[X_{N|k}], \mathbf{v}_{N|k}) \in \tilde{\mathcal{Z}}_N$ and thus (78) holds for $i = N - 1$ as well.

Furthermore, it must be proven that the terminal condition (58d) is fulfilled. Similarly to the iterative cost, it can be shown using Assumption 5 and Lemma 7 that

$$q_j(\mathbb{E}[X_{N-1|k+1}]) \leq q_j(\mathbb{E}[X_{N|k}]) + L_{q,j}(\hat{\delta}_N - \hat{\delta}_{N-1}) \quad (79)$$

holds. The sequence of parameters $\mathbf{v}_k$ is a feasible solution for the initial state $\mathbf{x}_k$ and thus, $\mathbb{E}[X_{N|k}] \in \mathbb{X}_f$. Inserting the upper bound for q_j from (54) in (79) results in

$$q_j(\mathbb{E}[X_{N-1|k+1}]) \leq L_{q,j}(\hat{\delta}_N - \hat{\delta}_{N-1} - \hat{\delta}_N) = -L_{q,j}\hat{\delta}_{N-1} < 0$$

and hence $\mathbb{E}[X_{N-1|k+1}] \in \mathbb{X}_f$ is guaranteed. From (56) follows that $\mathbb{E}[X_{N|k+1}] \in \mathbb{X}_f$ if $\hat{\mathbf{v}}$ is applied to the system. Thus, $\tilde{\mathbf{v}}_{k+1}$ is a feasible solution of $\mathbb{P}_N(\mathbf{x}_{k+1})$ and $\mathbf{x}_{k+1} \in \mathcal{X}_N$ if $\mathbf{x}_k \in \mathcal{X}_N$. $\square$

Figure 3 shows an example of the state prediction and constraint tightening for a two-dimensional system. For simplicity, the terminal constraint is assumed to be defined by linear functions

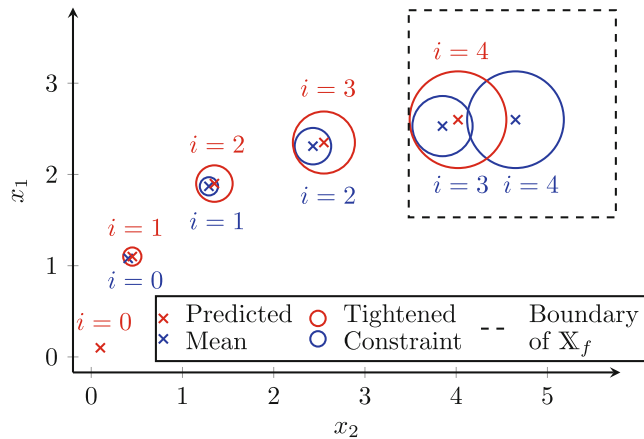


FIGURE 3 | Predicted mean and tightened constraint with the proposed controller at time step k (red) and with the shifted policy parameter sequence (69) at the next time step $k + 1$ (blue).

and thus, the terminal constraint is satisfied if the set of possible future states in the last prediction step $\mathbf{x}_{k+N}$ is a subset of $\mathbb{X}_f$. This set is shown as circle with the center at the mean of the predicted state and a radius $\hat{\delta}_i$. The red trajectory corresponds to the solution of $\mathbb{P}_N(\mathbf{x}_k)$ and thus fulfills the terminal constraint. The blue trajectory corresponds to the next time step $k + 1$ and assumes that the shifted policy parameter sequence (69) is applied to the system. Theorem 2 states that the set of possible future states in time step $k + 1$ for prediction step $i \in \mathbb{I}_{[0, N-1]}$ is a subset of the corresponding set in time step k for prediction step $i + 1$. The fulfillment of the terminal constraint in time step $k + 1$ is guaranteed by Assumption 6 because all possible states in prediction step $N - 1$ are an element of $\mathbb{X}_f$.

4.3 | Cost Decrease and Stability

The descent property is important in deterministic nonlinear MPC for guaranteeing asymptotic stability of the closed-loop system [1]. Under certain assumptions, a modified descent property can be stated for SMPC that bounds the expected value of the cost in the next time step for unconstrained systems [19] and for systems with bounded disturbances [20]. It is shown in the following that an equivalent modified descent property can be found for the proposed learning-based stochastic model predictive controller. Therefore, the disturbance sequence

$$\begin{aligned} \tilde{\mathbf{W}}_{k+1} &= (\mathbf{W}_{0|k+1}, \mathbf{W}_{1|k+1}, \dots, \mathbf{W}_{N-1|k+1}) \\ &:= (\mathbf{W}_{1|k}, \mathbf{W}_{2|k}, \dots, \mathbf{W}_{N-1|k}, \hat{\mathbf{W}}) \end{aligned} \quad (80)$$

is defined which is a shifted version of $\overline{\mathbf{W}}$, where $\hat{\mathbf{W}} \sim \mathcal{N}(\mathbf{0}, \Sigma(\mathbf{x}, \kappa_f(\mathbf{x})))$ with arbitrary $\mathbf{x} \in \mathbb{X}_f$ is added at the end.

Theorem 3. Let Assumptions 1–6 hold. Then, it holds that

$$\mathbb{E}[V_N^*(X_{1|k})] \leq V_N^*(\mathbf{x}_k) - \ell(\mathbf{x}_k, \kappa_N(\mathbf{x}_k)) + \epsilon \quad (81)$$

with ϵ from Assumption 6.

Proof. This proof of the modified descent property follows the corresponding proof in [20] and [19, Theorem 3] but adapts it regarding the problem statement of this paper. The control policy sequence $\tilde{\mathbf{v}}_{k+1}$ is feasible for $\mathbb{P}_N(\mathbf{x}_{k+1})$ as shown in the proof of Theorem 2. Since $\tilde{\mathbf{W}}_{k+1}$ is the shifted sequence $\overline{\mathbf{W}}_k$, the cost in the next time step depending on the disturbance sequence is given by

$$\begin{aligned} J_N(X_{1|k}, \tilde{\mathbf{v}}_{k+1}, \tilde{\mathbf{W}}_{k+1}) &= J_N(\mathbf{x}_k, \tilde{\mathbf{v}}_k^*, \overline{\mathbf{W}}_k) - \ell(\mathbf{x}_k, \kappa_N(\mathbf{x}_k)) - V_f(X_{N|k}) \\ &\quad + \ell(X_{N-1|k+1}, \kappa_f(X_{N-1|k+1})) + V_f(X_{N|k+1}). \end{aligned}$$

The predicted state $X_{N|k+1}$ is calculated with the prediction model (42) using the disturbance $\hat{\mathbf{W}}$ which leads to

$$\begin{aligned} J_N(X_{1|k}, \tilde{\mathbf{v}}_{k+1}, \tilde{\mathbf{W}}_{k+1}) &= J_N(\mathbf{x}_k, \tilde{\mathbf{v}}_k^*, \overline{\mathbf{W}}_k) - \ell(\mathbf{x}_k, \kappa_N(\mathbf{x}_k)) - V_f(X_{N|k}) \\ &\quad + \ell(X_{N|k}, \kappa_f(X_{N|k})) \\ &\quad + V_f(f_\pi(X_{N|k}, \hat{\mathbf{v}})) + \mu(X_{N|k}, \hat{\mathbf{v}}) + \hat{\mathbf{W}}. \end{aligned} \quad (82)$$

Taking expectation on both sides of (82) and applying (57) results in

$$\begin{aligned} & \mathbb{E}[J_N(X_{1|k}, \tilde{\mathbf{v}}_{k+1}, \tilde{\mathbf{W}}_{k+1})] \\ & \leq \mathbb{E}[J_N(\mathbf{x}_k, \tilde{\mathbf{v}}_k^*, \tilde{\mathbf{W}}_k)] - \ell(\mathbf{x}_k, \kappa_N(\mathbf{x}_k)) + \epsilon, \end{aligned} \quad (83)$$

where $\mathbb{E}[J_N(\mathbf{x}_k, \tilde{\mathbf{v}}_k^*, \tilde{\mathbf{W}}_k)]$ can be replaced with $V_N^*(\mathbf{x}_k)$. From the definition of V_N in (46), we have

$$\mathbb{E}[V_N(X_{1|k}, \tilde{\mathbf{v}}_{k+1})] = \mathbb{E}[J_N(X_{1|k}, \tilde{\mathbf{v}}_{k+1}, \tilde{\mathbf{W}}_{k+1})], \quad (84)$$

where the expected value on the left-hand side is required because $X_{1|k}$ is a random variable. Moreover, it must hold that

$$\mathbb{E}[V_N^*(X_{1|k})] \leq \mathbb{E}[V_N(X_{1|k}, \tilde{\mathbf{v}}_k)] \quad (85)$$

and thus it follows from (83–85)

$$\mathbb{E}[V_N^*(X_{1|k})] \leq V_N^*(\mathbf{x}_k) - \ell(\mathbf{x}_k, \kappa_N(\mathbf{x}_k)) + \epsilon \quad (86)$$

with ϵ from Assumption 6. $\square$

Finally, it can be shown that $(\mathbb{E}[\|X_{i|k}\|])$ is uniformly bounded for $i \geq 0$ if there exist functions $\alpha_\ell, \alpha_f \in \mathcal{K}_\infty$ that satisfy

$$\ell(\mathbf{x}, \mathbf{u}) \geq \alpha_\ell(\|\mathbf{x}\|) \quad \forall \mathbf{x} \in \mathbb{X}, \mathbf{u} \in \mathbb{U}, \quad (87)$$

$$V_f(\mathbf{x}) \leq \alpha_f(\|\mathbf{x}\|) \quad \forall \mathbf{x} \in \mathbb{X}_f, \quad (88)$$

see [20, Section 4] for details. This is a common definition of stochastic stability of a system.

5 | Numerical Evaluation

Consider the following two-dimensional nonlinear system

$$\begin{bmatrix} x_{1,k+1} \\ x_{2,k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} x_{1,k} + \frac{1}{2}(x_{2,k} + u_k) \\ x_{2,k} + u_k \end{bmatrix}}_{f(\mathbf{x}_k, u_k)} + \underbrace{\begin{bmatrix} 0.1 \tanh(x_{1,k}) \\ 0.1 \sin(u_k) \end{bmatrix}}_{g(\mathbf{x}_k, u_k)} \quad (89)$$

with state $\mathbf{x} \in \mathbb{R}^2$ and control input $u \in \mathbb{R}$. The part $f(\mathbf{x}_k, u_k)$ is considered to be known and the part $g(\mathbf{x}_k, u_k)$ is considered as unknown. For reasons of simplicity, we choose the control policy

$$u_k = \pi(\mathbf{x}_k, v_k) = v_k \quad (90)$$

such that optimizing over u_k is equivalent to optimizing over v_k . Thus, the resulting known and unknown parts of the system of interest are

$$\mathbf{f}_\pi(\mathbf{z}_k) = \begin{bmatrix} x_{1,k} + \frac{1}{2}(x_{2,k} + v_k) \\ x_{2,k} + v_k \end{bmatrix} \quad (91)$$

$$\mathbf{g}_\pi(\mathbf{z}_k) = \begin{bmatrix} 0.1 \tanh(x_{1,k}) \\ 0.1 \sin(v_k) \end{bmatrix} \quad (92)$$

with $\mathbf{z}_k = (\mathbf{x}_k, v_k)$. Furthermore, the constraints

$$h_1(x_{1,k}) = |x_{1,k}| - 3 \leq 0 \quad (93)$$

$$h_2(x_{2,k}) = |x_{2,k}| - 3 \leq 0 \quad (94)$$

$$h_3(v_k) = |v_k| - 2 \leq 0 \quad (95)$$

must be satisfied, which can be combined to the constraint $(\mathbf{x}, v) \in \mathcal{Z}$ with

$$\mathcal{Z} = \left\{ (\mathbf{x}, v) \in \mathbb{R}^2 \times \mathbb{R} \mid |x_1| \leq 3, |x_2| \leq 3, |v| \leq 2 \right\}. \quad (96)$$

The unknown function g_π should be approximated by two independent GPs

$$\tilde{g}_{\pi,1} \sim \mathcal{GP}(0, k_1), \quad \tilde{g}_{\pi,2} \sim \mathcal{GP}(0, k_2) \quad (97)$$

with kernels

$$\begin{aligned} k_1(x_1, x'_1) &= C_1 \exp(-\beta_1 |x_1 - x'_1|) \\ k_2(v, v') &= C_2 \exp(-\beta_2 |v - v'|), \end{aligned} \quad (98)$$

where $0 < C_1, C_2, \beta_1, \beta_2 < \infty$ are the parameters of the kernels. These define two RKHS, for which an upper bound of the corresponding RKHS norm for $g_{\pi,1}$ and $g_{\pi,2}$ must be known according to Assumption 1. It can be calculated by [31]

$$\begin{aligned} \|g_{\pi,i}\|_{\mathcal{H}_{k,i}}^2 &= \frac{1}{2C_i} (g_{\pi,i}(a_1)^2 + g_{\pi,i}(a_2)^2) \\ &+ \frac{1}{2\beta_i C_i} \int_{a_1}^{a_2} g'_{\pi,i}(t_i)^2 + \beta_i^2 g_{\pi,i}(t_i)^2 d\lambda(t_i) \end{aligned} \quad (99)$$

for $i \in \{1, 2\}$, where C_i and β_i are the kernel parameters in (98), a_1 and a_2 are the bounds of the corresponding Sobolev space, $t_1 = x_1$ and $t_2 = v$ are the variables of integration, $g'_{\pi,i}(t_i)$ is the derivative of $g_{\pi,i}(t_i)$, and λ is the corresponding Lebesgue measure. The norm (99) cannot be evaluated if $g_{\pi,i}$ is unknown. However, (99) is used in this example to verify that Assumption 1 is satisfied. In the following, we assume that the GPs (97) use M equidistant data points which fulfill Assumption 3.

The stochastic model predictive controller considers the terminal constraint $\mathbf{x}_{k+N} \in \mathbb{X}_f$ with prediction horizon $N = 4$ and terminal set

$$\mathbb{X}_f = \left\{ \mathbf{x} \in \mathbb{R}^2 \mid |x_1| \leq 1, |x_2| \leq 1 \right\}. \quad (100)$$

The stage cost and the terminal cost are defined by

$$\ell(\mathbf{x}, u) = x_1^2 + x_2^2 + u^2 \quad (101)$$

$$V_f(\mathbf{x}) = 2x_1^2 + 2x_2^2. \quad (102)$$

The terminal control law $\kappa_f(\mathbf{x})$ is designed by linearizing system (89) at the origin and constructing a linear-quadratic regulator for the linearized system. We confirmed simulatively the fulfillment of Assumption 6 with the control law κ_f .

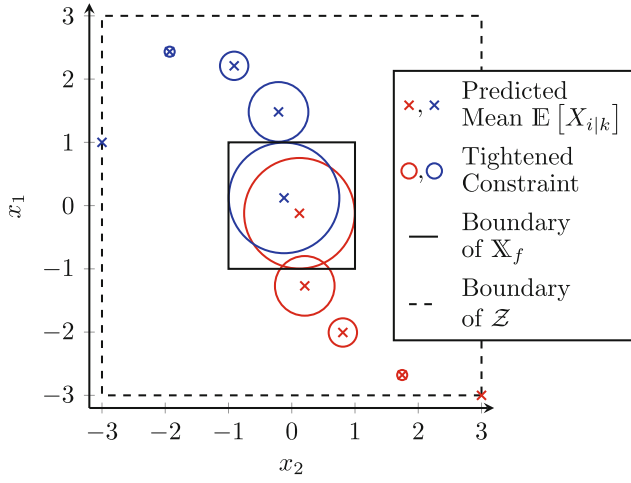


FIGURE 4 | Predicted mean and tightened constraint computed in the first time step by the proposed controller with $M = 10$ for the initial positions $\mathbf{x}_k = [3, -3]^T$ (red) and $\mathbf{x}_k = [-3, 1]^T$ (blue).

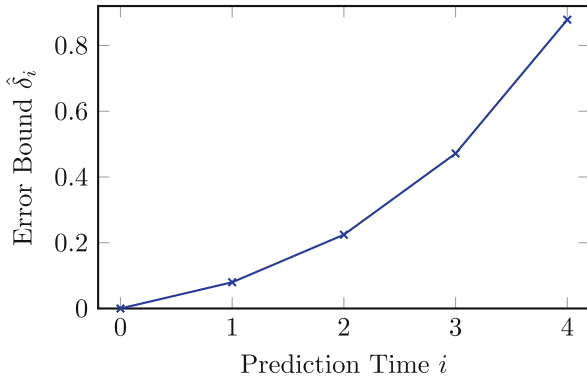


FIGURE 5 | Error bound $\hat{\delta}_i$ as a function of the prediction time according to (51).

In the following, the number of data points is set to $M = 10$ and the upper bounds b_1 and b_2 are set 50% larger than the values obtained by solving (99) because they are usually not available and must be approximated conservatively. Figure 4 shows the predicted expected value of the state and the tightened constraint for two different initial states. In both cases, the inequality constraints and the terminal inequality constraints are fulfilled. It can be seen that the error bound $\hat{\delta}_i$, which is equal to the radius of the circles in Figure 4, grows significantly with increasing prediction time i , as shown in Figure 5. This limits the prediction horizon of the controller to an upper bound because the terminal constraint cannot be satisfied if N is too large.

In Figure 6, the predicted mean trajectory and the corresponding error bound in time step k are plotted in red for the optimal solution of $\mathbb{P}_N(\mathbf{x}_k)$, which is identical to the red trajectory in Figure 4. It is compared to its counterpart in the next time step $k + 1$ in blue and the trajectory that results from the shifted policy parameter sequence (69) in green. The trajectory with the shifted policy parameter sequence is guaranteed to be a feasible solution of $\mathbb{P}_N(\mathbf{x}_{k+1})$ due to Theorem 2. It is not optimal in the next time step

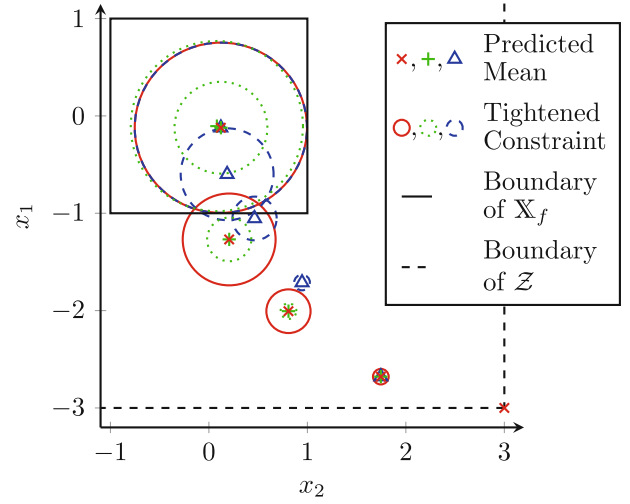


FIGURE 6 | Predicted mean and tightened constraint with the proposed controller at time step k (red), with the proposed controller at the next time step $k + 1$ (blue) and with the shifted policy parameter sequence (69) at time step $k + 1$ (green). The initial state is $\mathbf{x}_k = [3, -3]^T$.

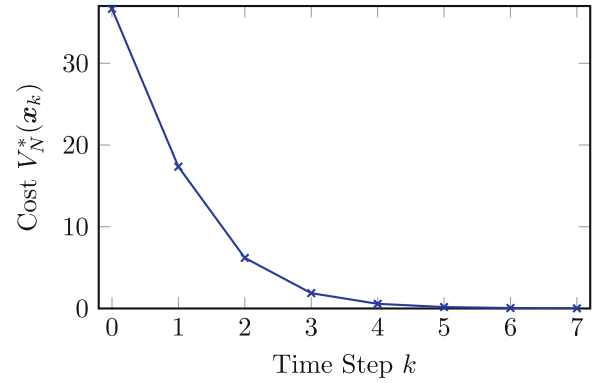


FIGURE 7 | Cost associated with the solution of $\mathbb{P}_N(\mathbf{x}_k)$ depending on the time step k .

but may be used as an initial solution in time step $k + 1$ for an iterative solver. Figure 7 shows the cost of the solution of $\mathbb{P}_N(\mathbf{x}_k)$ as a function of k . It can be observed that the cost decreases. However, the descent property is only guaranteed for the expected value of the cost according to Theorem 3.

6 | Conclusion

This article presents an SMPC approach for nonlinear systems with partially unknown dynamics, where the unknown part is approximated by a GP. An error bound of the predicted state is derived based on an upper bound of the RKHS norm of the unknown dynamics. In order to guarantee recursive feasibility, decrease of the expected cost, and stability, the constraints in the OCP, which is solved by the controller, are tightened using the error bound of the predicted state and Lipschitz constants of the inequality constraints. A terminal region and a corresponding stabilizing controller for this region is assumed, as it is common in the literature. Due to the possibly nonlinear dynamics and its unknown parts, the error bound can grow significantly with

increasing prediction time, as shown in the numerical evaluation. Thus, the tightening of the constraints can result in an initially infeasible OCP for large prediction horizons. This results in an upper bound of the prediction horizon for all non-trivial system dynamics. A perspective for future work could be the transfer of results to continuous-time systems.

Author Contributions

Daniel Landgraf: conceptualization, formal analysis, methodology, software, writing – original draft, visualization. **Andreas Völz:** conceptualization, writing – review and editing, supervision, project administration. **Knut Graichen:** conceptualization, writing – review and editing, supervision, project administration

Acknowledgments

Open Access funding enabled and organized by Projekt DEAL.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

References

1. D. Mayne, J. Rawlings, C. Rao, and P. Scokaert, “Constrained Model Predictive Control: Stability and Optimality,” *Automatica* 36, no. 6 (2000): 789–814.
2. J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. (Nob Hill Publishing, 2017).
3. G. Pannocchia, J. B. Rawlings, and S. J. Wright, “Conditions Under Which Suboptimal Nonlinear MPC Is Inherently Robust,” *Systems & Control Letters* 60, no. 9 (2011): 747–755.
4. L. Hewing, J. Kabzan, and M. N. Zeilinger, “Cautious Model Predictive Control Using Gaussian Process Regression,” *IEEE Transactions on Control Systems Technology* 28, no. 6 (2020): 2736–2743.
5. T. Koller, F. Berkenkamp, M. Turchetta, and A. Krause, “Learning-Based Model Predictive Control for Safe Exploration,” in *Proceedings of the IEEE Conference on Decision and Control (CDC)* (IEEE, 2018), 6059–6066.
6. Y. Pan and J. Wang, “Model Predictive Control of Unknown Nonlinear Dynamical Systems Based on Recurrent Neural Networks,” *IEEE Transactions on Industrial Electronics* 59, no. 8 (2012): 3089–3101.
7. Y. M. Ren, M. S. Alhajeri, J. Luo, et al., “A Tutorial Review of Neural Network Modeling Approaches for Model Predictive Control,” *Computers & Chemical Engineering* 165 (2022): 107956.
8. S. R. Chowdhury and A. Gopalan, “On Kernelized Multi-Armed Bandits,” in *Proceedings of the International Conference on Machine Learning (PMLR)*, vol. 70 (PMLR, 2017), 844–853.
9. N. Srinivas, A. Krause, S. M. Kakade, and M. W. Seeger, “Information-Theoretic Regret Bounds for Gaussian Process Optimization in the Bandit Setting,” *IEEE Transactions on Information Theory* 58, no. 5 (2012): 3250–3265.
10. J. Umlauft, L. Pöhler, and S. Hirche, “An Uncertainty-Based Control Lyapunov Approach for Control-Affine Systems Modeled by Gaussian Process,” *IEEE Control Systems Letters* 2, no. 3 (2018): 483–488.
11. A. Lederer, J. Umlauft, and S. Hirche, “Uniform Error Bounds for Gaussian Process Regression With Application to Safe Control,” *Advances in Neural Information Processing Systems* 32 (2019): 1–11.
12. M. Kanagawa, P. Hennig, D. Sejdinovic, and B. K. Sriperumbudur, “Gaussian Processes and Kernel Methods: A Review on Connections and Equivalences,” 2018, <https://arxiv.org/abs/1807.02582>.
13. A. Mesbah, “Stochastic Model Predictive Control: An Overview and Perspectives for Future Research,” *IEEE Control Systems Magazine* 36, no. 6 (2016): 30–44.
14. J. A. Paulson, S. Streif, and A. Mesbah, “Stability for Receding-Horizon Stochastic Model Predictive Control,” in *Proceedings of the American Control Conference (ACC)* (IEEE, 2015), 937–943.
15. M. Lorenzen, F. Dabbene, R. Tempo, and F. Allgöwer, “Constraint-Tightening and Stability in Stochastic Model Predictive Control,” *IEEE Transactions on Automatic Control* 62, no. 7 (2017): 3165–3177.
16. J. Berberich, J. Köhler, M. A. Müller, and F. Allgöwer, “Data-Driven Model Predictive Control With Stability and Robustness Guarantees,” *IEEE Transactions on Automatic Control* 66, no. 4 (2021): 1702–1717.
17. G. Pan, R. Ou, and T. Faulwasser, “Towards Data-Driven Stochastic Predictive Control,” *International Journal of Robust and Nonlinear Control* 35, no. 7 (2025): 2588–2610.
18. D. Landgraf, A. Völz, F. Berkel, K. Schmidt, T. Specker, and K. Graichen, “Probabilistic Prediction Methods for Nonlinear Systems With Application to Stochastic Model Predictive Control,” *Annual Reviews in Control* 56 (2023): 100905.
19. D. Chatterjee and J. Lygeros, “On Stability and Performance of Stochastic Predictive Control Techniques,” *IEEE Transactions on Automatic Control* 60, no. 2 (2015): 509–514.
20. D. Q. Mayne and P. Falugi, “Stabilizing Conditions for Model Predictive Control,” *International Journal of Robust and Nonlinear Control* 29, no. 4 (2019): 894–903.
21. W. Zheng and B. Zhu, “Control Lyapunov–Barrier Function Based Model Predictive Control for Stochastic Nonlinear Affine Systems,” *International Journal of Robust and Nonlinear Control* 34, no. 1 (2024): 91–113.
22. R. D. McAllister and J. B. Rawlings, “Nonlinear Stochastic Model Predictive Control: Existence, Measurability, and Stochastic Asymptotic Stability,” *IEEE Transactions on Automatic Control* 68, no. 3 (2023): 1524–1536.
23. R. D. McAllister and J. B. Rawlings, “The Stochastic Robustness of Nominal and Stochastic Model Predictive Control,” *IEEE Transactions on Automatic Control* 68, no. 10 (2023): 5810–5822.
24. J. Köhler and M. N. Zeilinger, “Predictive Control for Nonlinear Stochastic Systems: Closed-Loop Guarantees With Unbounded Noise,” *IEEE Transactions on Automatic Control* 70, no. 11 (2025): 7382–7397.
25. R. D. McAllister and J. B. Rawlings, “On the Inherent Distributional Robustness of Stochastic and Nominal Model Predictive Control,” *IEEE Transactions on Automatic Control* 69, no. 2 (2024): 741–754.
26. F. Oldewurtel, C. N. Jones, and M. Morari, “A Tractable Approximation of Chance Constrained Stochastic MPC Based on Affine Disturbance Feedback,” in *Proceedings of the IEEE Conference on Decision and Control (CDC)* (IEEE, 2008), 4731–4736.
27. F. Berkenkamp, R. Moriconi, A. P. Schoellig, and A. Krause, “Safe Learning of Regions of Attraction for Uncertain, Nonlinear Systems With Gaussian Processes,” in *Proceeding of the IEEE Conference on Decision and Control (CDC)* (IEEE, 2016), 4661–4666.

28. C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning* (MIT Press, 2005).
29. A. Girard, C. Rasmussen, J. Candela, and R. Murray-Smith, "Gaussian Process Priors With Uncertain Inputs Application to Multiple-Step Ahead Time Series Forecasting," in *Advances in Neural Information Processing Systems*, vol. 15 (NeurIPS papers, 2002).
30. M. D. Springer, *The Algebra of Random Variables* (Wiley, 1979).
31. A. Berlinet and C. Thomas-Agnan, *Reproducing Kernel Hilbert Spaces in Probability and Statistics* (Springer, 2004).